Roof Condition Assessment

Sample Business Park

Building Diagnostic Robotics

Detailed roof condition findings and repair planning recommendations.

Building Scanned
Sample Business Park

Location
123 ABC St, Brooklyn, NY

Test Date
May 13th, 2026 10:00 AM

On Site Engineers
BDR Field Team

Headline Findings

65	Good	6 years
Roof Condition Index	Visual Condition	Remaining Roof Life

MRR Recommendation: **Moderate Repair Needed**

Remaining Service Life: **6 years**

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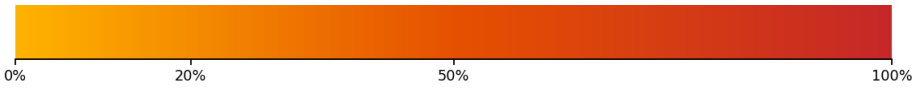
Executive Dashboard



Figure 1.1. Surveyed roof sections



Figure 1.2. All observed deficiencies



Below-Membrane Moisture Density Scale

1.1 Building Summary

Area Assessed: **5,483 sq.ft.**
Moisture: **4.16%**
Wet Area: **228 sq.ft.**
Risk Level: **Medium-Risk**
Roof Condition Index: **65**
Remaining Roof Life: **6 years**

1.2 Section Summary

Section	Area	Moisture	RCI	FCI	MCI	ICI	H/M/L	Condition	Repair	Life	MRR Recommendation
1	1,058	3.87%	75	86	84	71	11/29/10	Very Good	$4,375	8 yr	Minor Repair Needed
2	2,446	6.10%	35	35	32	49	52/70/66	Poor	$12,820	0 yr	Replacement Probable
3	1,979	1.93%	77	89	77	74	13/16/14	Very Good	$3,420	8 yr	Minor Repair Needed

Table 1.1. Building assessment summary by section

Capital Planning

Defect Type	Severity	Qty	Unit	Total Cost
Patching	Low	1,275	SF	$4,689.90
Improper Equipment Supports	Low	32	EA	$3,840.00
Ridges	Low	165	SF	$2,866.05
Base Flashing	High	80	SF	$2,009.00
Base Flashing	Medium	150	LF	$1,270.50
Base Flashing	Low	135	LF	$1,143.45
Patching	Medium	250	SF	$939.00
Ponding	Low	2	EA	$683.85
Metal Cap Flashing	High	24	SF	$602.70
Defective Seams	Medium	40	LF	$578.40
Metal Cap Flashing	Low	45	LF	$381.15
Metal Cap Flashing	Medium	40	LF	$338.80
Surface Deterioration	Low	50	SF	$329.00
Flashed Penetration	Medium	3	EA	$249.33
Defective Seams	Low	15	LF	$216.90
Debris Vegetation	Low	37	EA	$185.00
Splits	High	25	SF	$166.74
Flashed Penetration	High	1	EA	$83.11
Base Flashing Membrane Material	Medium	5	LF	$42.35
Total estimated repair cost (building-wide)				**$20,615.23**

Table 2.1. Estimated repair costs by severity (building-wide)

MRR Recommendation

Moderate Repair Needed

Remaining Service Life

5 years 10 months

2.1 Roof Condition & Service Life

Current roof age: **25 years**
Roof condition: **Good (RCI 65)**
Estimated remaining roof life: **5 years 10 months**
MRR recommendation: **Moderate Repair Needed**

2.2 Cost-to-ASL Analysis

2.2.1 Building-wide

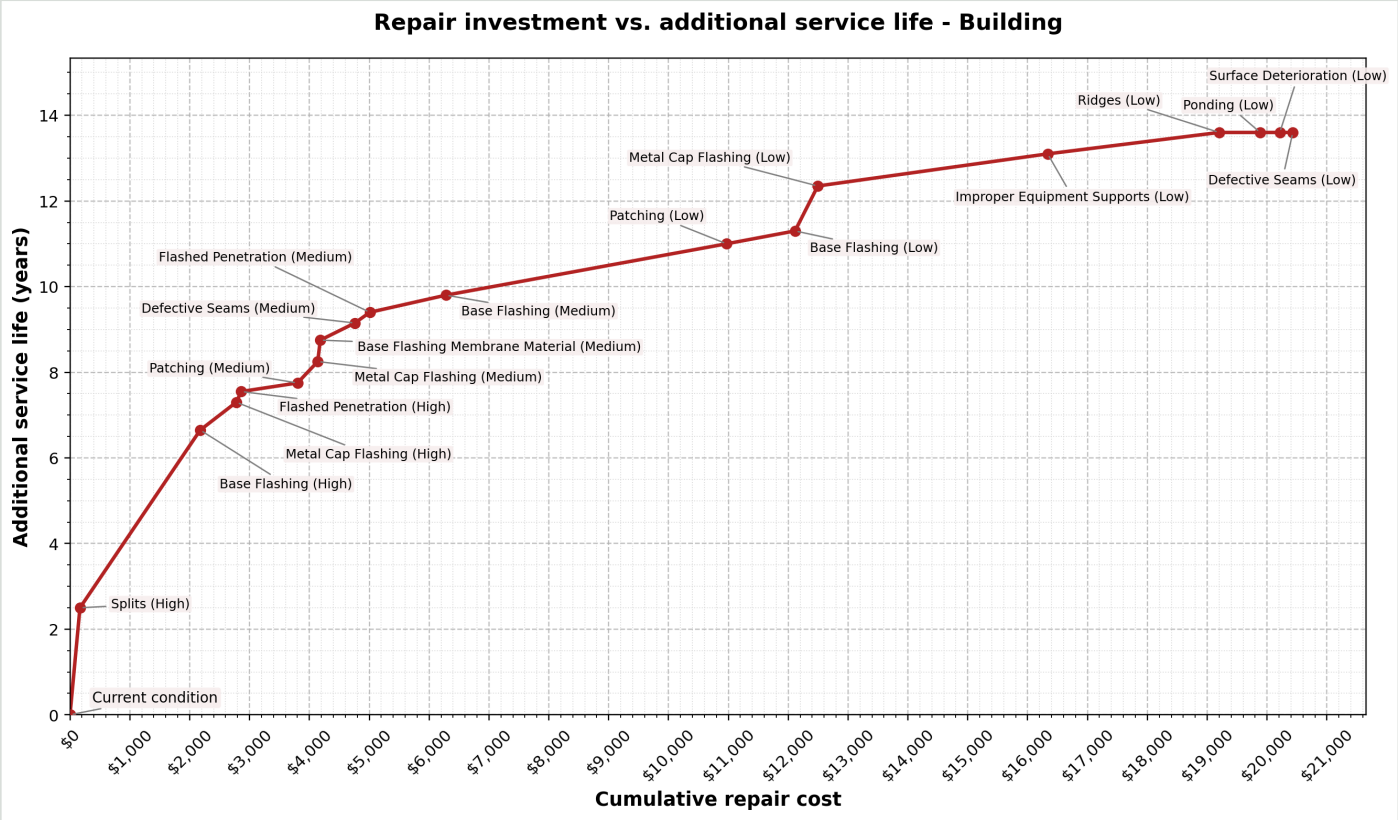


Figure 2.1. Building-wide repair investment vs. additional service life

Highest-return building-wide repairs

Defect Type	Severity	Repair Cost	Additional Service Life
Splits	High	$167	+2 years 6 months
Base Flashing	High	$2,009	+4 years 2 months
Metal Cap Flashing	Low	$381	+1 year 1 month
Metal Cap Flashing	Medium	$339	+7 months
Metal Cap Flashing	High	$603	+9 months

Table 2.2. Highest-return building-wide repairs by additional service life

Section 1

Area Assessed: **1,058 sq.ft.**
Current roof age: **25 years**
Estimated remaining roof life: **8 years**
Total Visual Defects: **50**
Roof condition: **Very Good (RCI 75)**
MRR recommendation: **Minor Repair Needed**

3.1 Cost-to-ASL Analysis

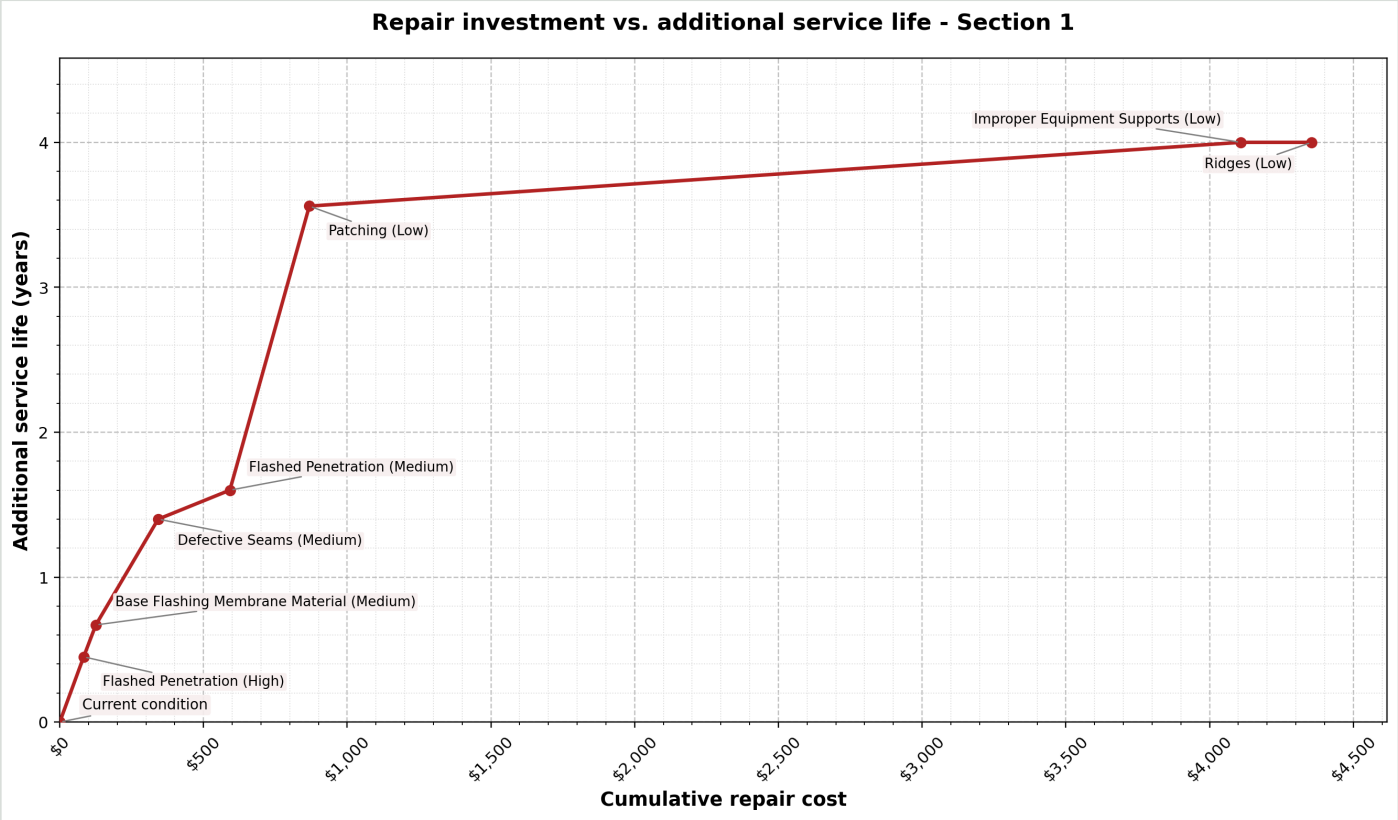


Figure 3.1. Section 1 repair investment vs. additional service life

Highest-return repairs (Section 1)

Defect Type	Severity	Repair Cost	Additional Service Life
Patching	Low	$277	+2 years
Flashed Penetration	Medium	$249	+9 months

Table 3.1. Highest-return repairs (Section 1) by additional service life

3.2 Maps and Condition



Figure 3.2. Below-membrane moisture



Figure 3.3. Visual defects

Metric	Value
Visual Condition Rating	Very Good
Moisture	3.87%
Roof Condition Index (RCI)	75
Membrane Condition Index (MCI)	84
Flashing Condition Index (FCI)	86
Insulation Condition Index (ICI)	71
MRR Recommendation	Minor Repair Needed

Table 3.2. Section 1 condition summary

Color	Defect Type	Count
	Improper Equipment Supports	29
	Defective Seams	5
	Patching	5
	Debris And Vegetation	4
	Flashed Penetrations	4
	Ridges	2
	Base Flashing-Membrane Material	1

Table 3.3. Section 1 defect inventory

3.3 Repair Scope

Defect Type	Severity	Qty	Unit	Total
Improper Equipment Supports	Low	27	EA	$3,240.00
Patching	Low	100	SF	$276.60
Flashed Penetration	Medium	3	EA	$249.33
Ridges	Low	15	SF	$247.05
Defective Seams	Medium	15	LF	$216.90
Flashed Penetration	High	1	EA	$83.11
Base Flashing Membrane Material	Medium	5	LF	$42.35
Debris Vegetation	Low	4	EA	$20.00
Section repair subtotal				**$4,375.34**

Table 3.4. Repair scope for Section 1

Section 2

Area Assessed: **2,446 sq.ft.**
Total Visual Defects: **188**
Current roof age: **25 years**
Roof condition: **Poor (RCI 35)**
Estimated remaining roof life: **0 years**
MRR recommendation: **Replacement Probable**

4.1 Cost-to-ASL Analysis

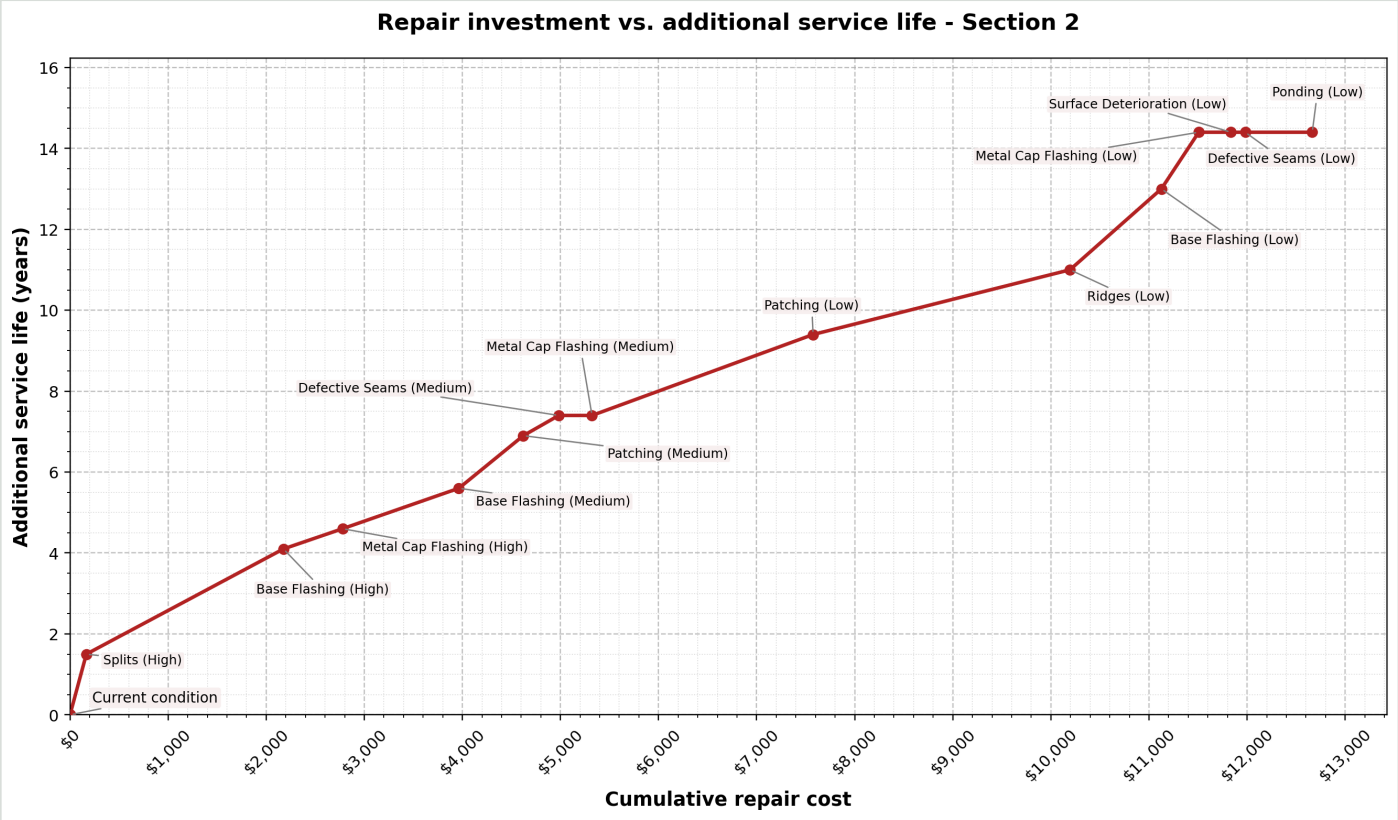


Figure 4.1. Section 2 repair investment vs. additional service life

Highest-return repairs (Section 2)

Defect Type	Severity	Repair Cost	Additional Service Life
Splits	High	$167	+1 year 6 months
Metal Cap Flashing	Low	$381	+1 year 5 months
Base Flashing	Low	$932	+2 years
Base Flashing	High	$2,009	+2 years 7 months
Patching	Medium	$657	+1 year 3 months

Table 4.1. Highest-return repairs (Section 2) by additional service life

4.2 Maps and Condition

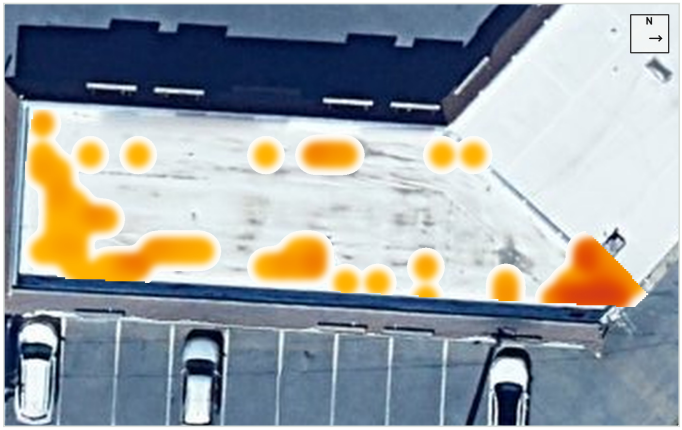


Figure 4.2. Below-membrane moisture

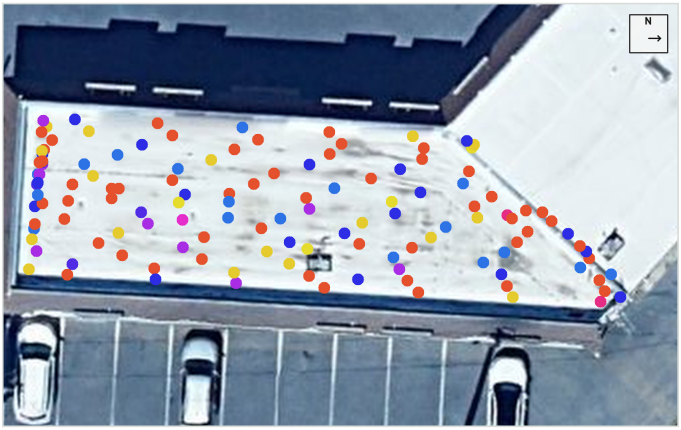


Figure 4.3. Visual defects

Metric	Value
Visual Condition Rating	Poor
Moisture	6.10%
Roof Condition Index (RCI)	35
Membrane Condition Index (MCI)	32
Flashing Condition Index (FCI)	35
Insulation Condition Index (ICI)	49
MRR Recommendation	Replacement Probable

Table 4.2. Section 2 condition summary

Color	Defect Type	Count
	Base Flashing	73
	Debris And Vegetation	33
	Patching	33
	Metal Cap Flashing	23
	Ridges	12
	Defective Seams	8
	Ponding	3
	Surface Deterioration	2
	Splits	1

Table 4.3. Section 2 defect inventory

4.3 Repair Scope

Defect Type	Severity	Qty	Unit	Total
Ridges	Low	150	SF	$2,619.00
Patching	Low	600	SF	$2,253.60
Base Flashing	High	80	SF	$2,009.00
Base Flashing	Medium	140	LF	$1,185.80
Base Flashing	Low	110	LF	$931.70
Ponding	Low	2	EA	$683.85
Patching	Medium	175	SF	$657.30
Metal Cap Flashing	High	24	SF	$602.70
Metal Cap Flashing	Low	45	LF	$381.15
Defective Seams	Medium	25	LF	$361.50
Metal Cap Flashing	Medium	40	LF	$338.80
Surface Deterioration	Low	50	SF	$329.00
Splits	High	25	SF	$166.74
Debris Vegetation	Low	31	EA	$155.00
Defective Seams	Low	10	LF	$144.60
Section repair subtotal				**$12,819.74**

Table 4.4. Repair scope for Section 2

Section 3

Area Assessed: **1,979 sq.ft.**
Total Visual Defects: **43**
Current roof age: **25 years**
Roof condition: **Very Good (RCI 77)**
Estimated remaining roof life: **8 years**
MRR recommendation: **Minor Repair Needed**

5.1 Cost-to-ASL Analysis

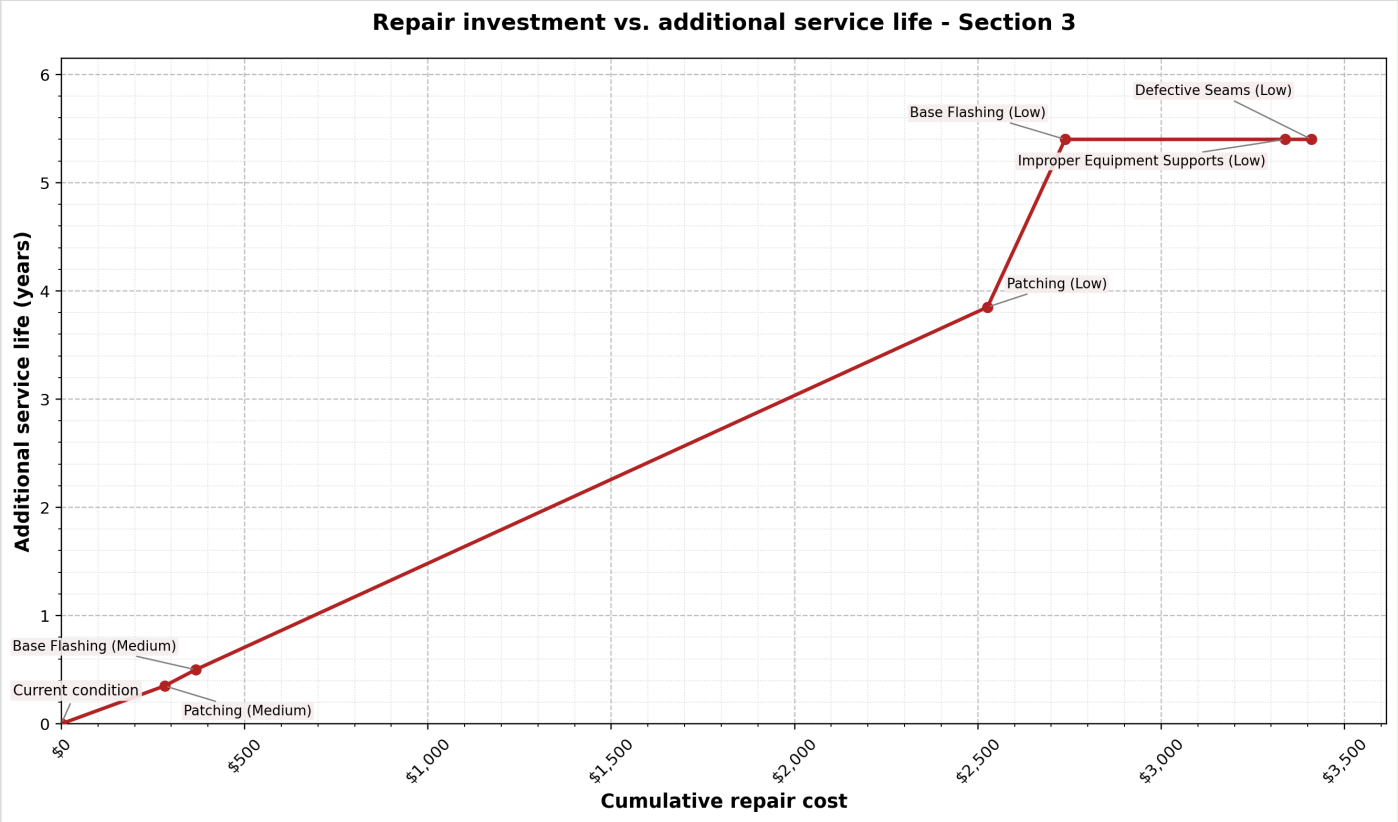


Figure 5.1. Section 3 repair investment vs. additional service life

Highest-return repairs (Section 3)

Defect Type	Severity	Repair Cost	Additional Service Life
Base Flashing	Low	$212	+1 year 6 months
Patching	Low	$2,160	+3 years 4 months

Table 5.1. Highest-return repairs (Section 3) by additional service life

5.2 Maps and Condition

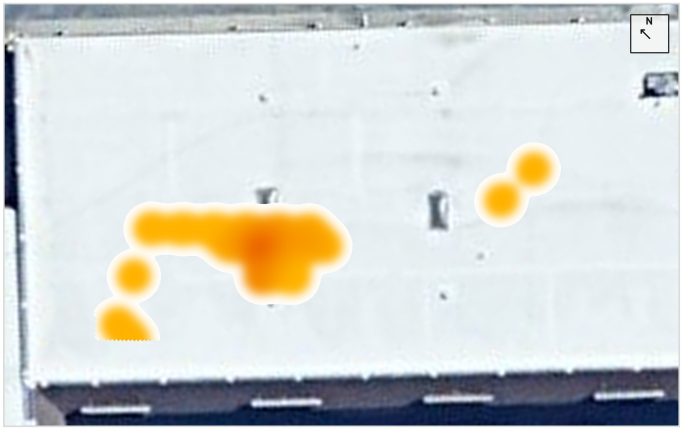


Figure 5.2. Below-membrane moisture

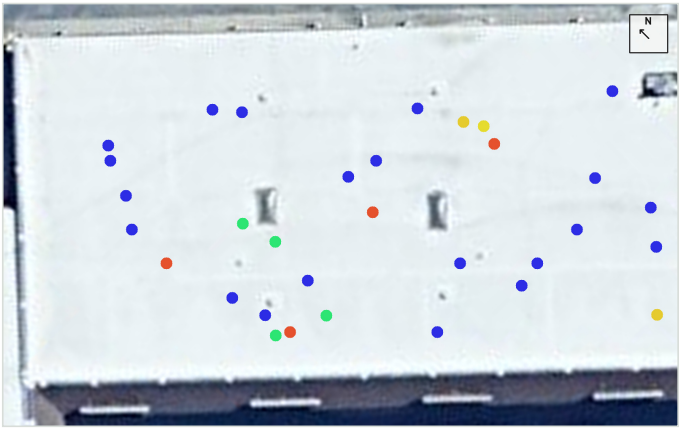


Figure 5.3. Visual defects

Metric	Value
Visual Condition Rating	Very Good
Moisture	1.93%
Roof Condition Index (RCI)	77
Membrane Condition Index (MCI)	77
Flashing Condition Index (FCI)	89
Insulation Condition Index (ICI)	74
MRR Recommendation	Minor Repair Needed

Table 5.2. Section 3 condition summary

Color	Defect Type	Count
	Patching	28
	Base Flashing	7
	Improper Equipment Supports	5
	Debris And Vegetation	2
	Defective Seams	1

Table 5.3. Section 3 defect inventory

5.3 Repair Scope

Defect Type	Severity	Qty	Unit	Total
Patching	Low	575	SF	$2,159.70
Improper Equipment Supports	Low	5	EA	$600.00
Patching	Medium	75	SF	$281.70
Base Flashing	Low	25	LF	$211.75
Base Flashing	Medium	10	LF	$84.70
Defective Seams	Low	5	LF	$72.30
Debris Vegetation	Low	2	EA	$10.00
Section repair subtotal				**$3,420.15**

Table 5.4. Repair scope for Section 3

Project Context

6.1 Site and Survey

Metric	Details
Facility	Sample Business Park
Location	123 ABC St, Brooklyn, NY
Survey date	May 13th, 2026 10:00 AM

6.2 Assessment Scope

Metric	Summary
Roof sections surveyed	3
Assessed scan area	5,483 sq.ft.
Survey methods	GPR moisture mapping and geolocated visual defect documentation

6.3 Conditions at Survey

Metric	Reading
Temperature	70.5°F
Wind	6.7 mph from the North-Northeast
Humidity	40.3%
Cloud cover	0.3%
Pressure	1008.0 inHg
Recent rainfall	No significant rainfall in 48 hours prior to scan